

Bifurcation analysis and chaos control in discrete-time glycolysis models

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Abstract In this paper, the qualitative behavior of two discrete-time glycolysis models is discussed. The discrete-time models are obtained by implementing forward Euler's scheme and nonstandard finite difference method. The parametric conditions for local asymptotic stability of positive steady-states are investigated. Moreover, we discuss the existence and directions of period-doubling and Neimark–Sacker bifurcations with the help of center manifold theorem and bifurcation theory. OGY feedback control and hybrid control methods are implemented in order to control chaos in discrete-time glycolysis model due to emergence of period-doubling and Neimark–Sacker bifurcations. Numerical simulations are provided to illustrate theoretical discussion.

Keywords Glycolysis model · Local stability · Period-doubling bifurcation · Neimark–Sacker bifurcation · Chaos control

1 Introduction

The conversion of glucose into pyruvate through metabolic process is known as glycolysis, and during this process usually two molecules of pyruvate are created at the cost of every molecule of glucose. Kerb cycle is completed in the process of the utilization of pyruvate molecules. In other words, glycolysis is a biochemical reaction in living cells in which these cells obtain energy by utilization of glucose, and this biochemical process can be modeled through the following differential system [1,2]:

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$$\begin{aligned}\frac{dx}{dt} &= \alpha y - x + x^2 y, \\ \frac{dy}{dt} &= \beta - \alpha y - x^2 y,\end{aligned}\tag{1.1}$$

where x and y denote concentrations (in dimensionless forms) of adenosine diphosphate (ADP) and fructose-6-phosphate (F6P), respectively, and $\alpha \geq 0$, $\beta \geq 0$ represent kinetic parameters. For certain parametric values of α and β , a stable limit cycle can be observed for system (1.1). Moreover, existence of this stable limit cycle guarantees that biochemical reaction attains its required stage [3]. For further detail related to glycolysis process and its mathematical modeling, we refer to [4–7]. Moreover, in [8–11] qualitative properties and bifurcation analysis of glycolysis models are discussed. Mickens [14] constructed a nonstandard difference scheme for the system (1.1), and proved positivity of solutions along with limit-cycle behavior of discrete-time model.

Next, the implementation of forward Euler's scheme gives the following discrete-time counterpart of (1.1):

$$\begin{aligned}x_{n+1} &= x_n + h \left(\alpha y_n - x_n + x_n^2 y_n \right), \\ y_{n+1} &= y_n + h \left(\beta - \alpha y_n - x_n^2 y_n \right),\end{aligned}\tag{1.2}$$

where h denotes step size for Euler's scheme. For similar type of discretization, we refer to [12, 13].

Furthermore, arguing as in [14], we have the following nonstandard discrete-time model corresponding to system (1.2):

$$\begin{aligned}x_{n+1} &= \frac{x_n + \alpha h y_n + 2h x_n^2 y_n}{1 + h + h x_n y_n}, \\ y_{n+1} &= \frac{\beta h + y_n}{1 + \alpha h + h x_{n+1}^2}.\end{aligned}\tag{1.3}$$

The aim of this paper is to study the local asymptotic stability of unique positive steady-state of (1.2) and (1.3). Moreover, the parametric conditions for existence and directions of period-doubling bifurcation and Neimark–Sacker bifurcation for positive equilibrium of system (1.2) are investigated. Furthermore, OGY and hybrid control strategies are implemented for chaos control due to emergence of period-doubling and Neimark–Sacker bifurcations.

2 Local stability of positive equilibrium

The equilibrium point of (1.2) can be obtained by solving the following system:

$$x = x + h \left(\alpha y - x + x^2 y \right), \quad y = y + h \left(\beta - \alpha y - x^2 y \right).\tag{2.1}$$

Then, it is easy to see that $P = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ is the unique positive equilibrium point of system (1.2). Furthermore, Jacobian matrix of (1.2) evaluated at P is given by:

$$J(P) = \begin{bmatrix} 1 - h + \frac{2h\beta^2}{\alpha + \beta^2} & h(\alpha + \beta^2) \\ -\frac{2h\beta^2}{\alpha + \beta^2} & 1 - h(\alpha + \beta^2) \end{bmatrix}.$$

Moreover, the characteristic polynomial of $J(P)$ is given by:

$$F(\lambda) = \lambda^2 - \left(2 - h \left(\alpha - 1 + \beta^2 + \frac{2\alpha}{\alpha + \beta^2} \right) \right) \lambda + 1 + h \left(1 + (h - 1)\beta^2 + \alpha \left(h - 1 - \frac{2}{\alpha + \beta^2} \right) \right). \quad (2.2)$$

Then, by simple computations it follows that

$$F(1) = h^2(\alpha + \beta^2) > 0, \quad (2.3)$$

$$F(-1) = 4 + h \left(2 + (h - 2)\beta^2 + \alpha \left(h - 2 - \frac{4}{\alpha + \beta^2} \right) \right), \quad (2.4)$$

and

$$F(0) = 1 + h \left(1 + (h - 1)\beta^2 + \alpha \left(h - 1 - \frac{2}{\alpha + \beta^2} \right) \right). \quad (2.5)$$

Arguing as in [26] (see also [24, 25, 27–29]) and taking into account the relations between roots and coefficients of a quadratic equation, we have the following result in order to discuss the dynamics of unique positive $\left(\beta, \frac{\beta}{\beta + \alpha^2} \right)$ of system (1.2).

Lemma 2.1 Assume that $\mathbb{F}(\lambda) = \lambda^2 - A\lambda + B$, and $\mathbb{F}(1) > 0$ with λ_1, λ_2 are roots of $\mathbb{F}(\lambda) = 0$. Then the following results hold true:

- (i) $|\lambda_1| < 1$ and $|\lambda_2| < 1$ if and only if $\mathbb{F}(-1) > 0$ and $\mathbb{F}(0) < 1$.
- (ii) $|\lambda_1| < 1$ and $|\lambda_2| > 1$, or $|\lambda_1| > 1$ and $|\lambda_2| < 1$ if and only if $\mathbb{F}(-1) < 0$.
- (iii) $|\lambda_1| > 1$ and $|\lambda_2| > 1$ if and only if $\mathbb{F}(-1) > 0$ and $\mathbb{F}(0) > 1$.
- (iv) $\lambda_1 = -1$ and $|\lambda_2| \neq 1$ if and only if $\mathbb{F}(-1) = 0$ and $\mathbb{F}(0) \neq \pm 1$.
- (v) λ_1 and λ_2 are conjugate complex numbers with $|\lambda_1| = 1$ and $|\lambda_2| = 1$ if and only if $A^2 - 4B < 0$ and $\mathbb{F}(0) = 1$.

Next, we assume that λ_1 and λ_2 be roots of (2.2), then unique positive equilibrium point $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ of (1.2) is called a sink if $|\lambda_1| < 1$ and $|\lambda_2| < 1$ and therefore a sink is always locally asymptotic stable equilibrium point. P is known as source or repeller if $|\lambda_1| > 1$ and $|\lambda_2| > 1$ and thus a source is always unstable. P is called a saddle point if $|\lambda_1| < 1$ and $|\lambda_2| > 1$, or $|\lambda_1| > 1$ and $|\lambda_2| < 1$ and thus a saddle point is also unstable. P is known as non-hyperbolic if either $|\lambda_1| = 1$ or $|\lambda_2| = 1$.

Furthermore, from (2.3) it follows that $F(1) = h^2(\alpha + \beta^2) > 0$. Therefore, one can apply Lemma 2.1 to prove the following results.

Lemma 2.2 For unique positive equilibrium point $P = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ of system (1.2) the following holds true:

(i) $\left(\beta, \frac{\beta}{\alpha + \beta^2}\right)$ is a sink if and only if

$$2h\beta^2 + 2h\alpha \left(1 + \frac{2}{\alpha + \beta^2}\right) < 4 + 2h + h^2(\alpha + \beta^2),$$

and

$$1 + h\beta^2 + \alpha h < \beta^2 + \alpha \left(1 + \frac{2}{\alpha + \beta^2}\right).$$

(ii) $\left(\beta, \frac{\beta}{\alpha + \beta^2}\right)$ is a saddle point if and only if

$$2h\beta^2 + 2h\alpha \left(1 + \frac{2}{\alpha + \beta^2}\right) > 4 + 2h + h^2(\alpha + \beta^2).$$

(iii) $\left(\beta, \frac{\beta}{\alpha + \beta^2}\right)$ is a repeller if and only if

$$2h\beta^2 + 2h\alpha \left(1 + \frac{2}{\alpha + \beta^2}\right) < 4 + 2h + h^2(\alpha + \beta^2),$$

and

$$1 + h\beta^2 + \alpha h > \beta^2 + \alpha \left(1 + \frac{2}{\alpha + \beta^2}\right).$$

(iv) Assume that λ_1 and λ_2 be roots of (2.2), then $\lambda_1 = -1$ and $|\lambda_2| \neq 1$ if and only if

$$2h\beta^2 + 2h\alpha \left(1 + \frac{2}{\alpha + \beta^2}\right) = 4 + 2h + h^2(\alpha + \beta^2),$$

$$2 + h \left(1 + (h-1)\beta^2 + \alpha \left(h-1 - \frac{2}{\alpha + \beta^2}\right)\right) \neq 0,$$

and

$$1 + (h-1)\beta^2 + \alpha \left(h-1 - \frac{2}{\alpha + \beta^2}\right) \neq 0.$$

(v) The roots of (2.2) are complex with modulus one if and only if

$$(\alpha - 1)^2 + 2(\alpha - 3)\beta^2 + \beta^4 + \frac{4\alpha^2}{(\alpha + \beta^2)^2} - \frac{4\alpha}{\alpha + \beta^2} < 0,$$

and

$$1 + (h - 1)\beta^2 + \alpha \left(h - 1 - \frac{2}{\alpha + \beta^2} \right) = 0.$$

Next, we discuss the dynamics of model (1.3) at its unique positive equilibrium point $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$. The variational matrix of the system (1.3) evaluated at $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ is given by:

$$V \left(\beta, \frac{\beta}{\alpha + \beta^2} \right) = \begin{bmatrix} \frac{\alpha + \beta^2 + 3h\beta^2}{\alpha + h\alpha + \beta^2 + 2h\beta^2} & \frac{h(\alpha + \beta^2)^2}{\alpha + h\alpha + \beta^2 + 2h\beta^2} \\ -\frac{2h\beta^2(\alpha + (1+3h)\beta^2)}{(\alpha + \beta^2)(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1+h(\alpha + \beta^2))} & \frac{(1+h)\alpha + \beta^2 - 2h(-1+h\alpha)\beta^2 - 2h^2\beta^4}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1+h(\alpha + \beta^2))} \end{bmatrix}.$$

The characteristic polynomial of $V \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ is given by

$$T(\lambda) = \lambda^2 - \Delta(\alpha, \beta, h)\lambda + \frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))}. \quad (2.6)$$

where

$$\Delta(\alpha, \beta, h) := \frac{\alpha(2 + h + h\alpha) + (2 + 5h + h(2 + h)\alpha)\beta^2 + h(1 + h)\beta^4}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))}.$$

Then from (2.6), it follows that

$$\begin{aligned} T(1) &= \frac{h^2(\alpha + \beta^2)^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} > 0, \\ T(-1) &= \frac{(2 + h)\alpha(2 + h\alpha) + 2(2 + 5h + 2h(1 + h)\alpha)\beta^2 + h(2 + 3h)\beta^4}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} > 0, \end{aligned}$$

and

$$T(0) = \frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} > 0.$$

Since $T(-1) > 0$, so it follows that there is no chance of period-doubling bifurcation for the system (1.3) at its unique positive equilibrium point $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$.

The following Lemma gives the dynamics of the system (1.3) related to its unique positive equilibrium point $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$.

Lemma 2.3 For unique positive equilibrium point $\left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ of system (1.2) the following results are satisfied:

(i) *Equilibrium point $\left(\beta, \frac{\beta}{\alpha+\beta^2}\right)$ of (1.3) is a sink if and only if*

$$\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} < 1.$$

(ii) *The unique equilibrium point $\left(\beta, \frac{\beta}{\alpha+\beta^2}\right)$ of system (1.3) cannot be a saddle point for all $\alpha, \beta > 0$ and $0 < h < 1$.*

(iii) *$\left(\beta, \frac{\beta}{\alpha+\beta^2}\right)$ is a source if and only if*

$$\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} > 1.$$

(iv) *The roots of (2.6) are complex with modulus one if and only if*

$$\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} = 1,$$

and

$$(\Delta(\alpha, \beta, h))^2 < 4 \left(\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} \right).$$

3 Period-doubling bifurcation

In this section, the parametric conditions for existence of period-doubling bifurcation of unique positive equilibrium point of the system (1.2) are discussed. For simplicity we take $k := \alpha + \beta^2$, then characteristic polynomial of the Jacobian matrix of (1.2) evaluated at unique positive equilibrium point $\left(\beta, \frac{\beta}{k}\right)$ can be rewritten as follows:

$$\mathbb{P}(\lambda) = \lambda^2 - \left(2 - h \left(k - 1 + \frac{2\alpha}{k}\right)\right) \lambda + \frac{(1 - hk)(k + hk - 2h\alpha)}{k} + 2h^2\beta. \quad (3.1)$$

Next, we assume that the following condition holds true:

$$(k + k^2 - 2\alpha)^2 - 8k^2\beta > 0. \quad (3.2)$$

Then, λ_1 and λ_2 be distinct real roots of (3.1). Moreover, we assume that

$$h = \frac{2k - 2k^2 - 4\alpha + \sqrt{(-2k + 2k^2 + 4\alpha)^2 + 16k(k^2 - 2k\alpha - 2k\beta)}}{2(k^2 - 2k\alpha - 2k\beta)}. \quad (3.3)$$

Then, roots of equation $\mathbb{P}(\lambda) = 0$ are $\lambda_1 = -1$ and

$$\lambda_2 = \frac{k - k^2 - 2\alpha + \sqrt{(k + k^2 - 2\alpha)^2 - 8k^2\beta}}{k(k - 2(\alpha + \beta))}.$$

Furthermore, $|\lambda_2| \neq 1$ under the following condition

$$\frac{(1 - hk)(k + hk - 2h\alpha)}{k} + 2h^2\beta \neq \pm 1. \quad (3.4)$$

Let us consider

$$\Omega_{FB} = \left\{ (\alpha, \beta, h) \in \mathbb{R}_+^3 : (3.2), (3.3) \text{ and } (3.4) \text{ are satisfied} \right\}.$$

In order to discuss the period-doubling bifurcation for the system (1.2) at its unique positive equilibrium point $\left(\beta, \frac{\beta}{k}\right)$, we take h as bifurcation parameter. Then, variation of parameters α , β and h in small neighborhood of Ω_{FB} gives emergence of period-doubling bifurcation. Furthermore, we set

$$h_1 = \frac{2k - 2k^2 - 4\alpha + \sqrt{(-2k + 2k^2 + 4\alpha)^2 + 16k(k^2 - 2k\alpha - 2k\beta)}}{2(k^2 - 2k\alpha - 2k\beta)}. \quad (3.5)$$

Then, for $(\alpha, \beta, h_1) \in \Omega_{FB}$, system (1.2) can be expressed by the following two-dimensional map:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \rightarrow \begin{pmatrix} X + h_1(\alpha Y - X + X^2 Y) \\ Y + h_1(\beta - \alpha Y - X^2 Y) \end{pmatrix}. \quad (3.6)$$

Assume that $\bar{h}$ be a small bifurcation parameter such that $|\bar{h}| \ll 1$, then corresponding perturbed map for (3.6) is given by:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \rightarrow \begin{pmatrix} X + (h_1 + \bar{h})(\alpha Y - X + X^2 Y) \\ Y + (h_1 + \bar{h})(\beta - \alpha Y - X^2 Y) \end{pmatrix}. \quad (3.7)$$

Then, map (3.7) has unique positive fixed point $\left(\beta, \frac{\beta}{k}\right)$. In order to translate this positive fixed point at origin $(0, 0)$, we consider the transformations $x = X - \beta$ and $y = Y - \frac{\beta}{k}$, then we get the following map:

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} g_1(x, y, \bar{h}) \\ g_2(x, y, \bar{h}) \end{pmatrix}, \quad (3.8)$$

where

$$g_1(x, y, \bar{h}) = a_{13}x^2 + a_{14}xy + b_1x^2y + c_1x\bar{h} + c_2y\bar{h} + c_3x^2\bar{h} + c_4xy\bar{h} + O\left((|x| + |y| + |\bar{h}|)^4\right),$$

$$g_2(x, y, \bar{h}) = a_{23}x^2 + a_{24}xy + d_1x^2y + e_1x\bar{h} + e_2y\bar{h} + e_3x^2\bar{h} + e_4xy\bar{h} + O\left((|x| + |y| + |\bar{h}|)^4\right),$$

$$\begin{aligned} a_{11} &= 1 + h_1 \left(\frac{2\beta^2}{k} - 1 \right), \quad a_{12} = h_1k, \quad a_{21} = -\frac{2h_1\beta^2}{k}, \quad a_{22} = 1 - h_1k, \\ a_{13} &= \frac{h_1\beta}{k}, \quad a_{14} = 2h_1\beta, \quad a_{23} = -\frac{h_1\beta}{k}, \quad a_{24} = -2h_1\beta, \end{aligned} \quad (3.9)$$

$$\begin{aligned} b_1 &= h_1, \quad c_1 = \frac{2\beta^2}{k} - 1, \quad c_2 = k, \quad c_3 = \frac{\beta}{k}, \quad c_4 = 2\beta, \\ d_1 &= -h_1, \quad e_1 = -\frac{2\beta^2}{k}, \quad e_2 = -k, \quad e_3 = -\frac{\beta}{k}, \quad e_4 = -2\beta. \end{aligned}$$

In order to convert the coefficient matrix $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ in map (3.8) into normal form, we use the following translation

$$\begin{pmatrix} x \\ y \end{pmatrix} = T \begin{pmatrix} u \\ v \end{pmatrix}, \quad (3.10)$$

where

$$T = \begin{pmatrix} a_{12} & a_{12} \\ -1 - a_{11} & \lambda_2 - a_{11} \end{pmatrix}$$

be an invertible matrix. From (3.8) and (3.10), we obtain

$$\begin{pmatrix} u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} -1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} g_3(u, v, \bar{h}) \\ g_4(u, v, \bar{h}) \end{pmatrix}, \quad (3.11)$$

where

$$\begin{aligned}
 g_3(u, v, \bar{h}) &= \left(\frac{a_{13}(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{a_{23}}{1 + \lambda_2} \right) x^2 + \left(\frac{a_{14}(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{a_{24}}{1 + \lambda_2} \right) xy \\
 &\quad + \left(\frac{b_1(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{d_1}{1 + \lambda_2} \right) x^2 y + \left(\frac{c_1(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{e_1}{1 + \lambda_2} \right) x \bar{h} \\
 &\quad + \left(\frac{c_2(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{e_2}{1 + \lambda_2} \right) y \bar{h} + \left(\frac{c_3(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{e_3}{1 + \lambda_2} \right) x^2 \bar{h} \\
 &\quad + \left(\frac{c_4(\lambda_2 - a_{11})}{a_{12}(1 + \lambda_2)} - \frac{e_4}{1 + \lambda_2} \right) xy \bar{h} + O\left((|u| + |v| + |\bar{h}|)^4\right), \\
 g_4(u, v, \bar{h}) &= \left(\frac{a_{13}(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{a_{23}}{1 + \lambda_2} \right) x^2 + \left(\frac{a_{14}(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{a_{24}}{1 + \lambda_2} \right) xy \\
 &\quad + \left(\frac{b_1(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{d_1}{1 + \lambda_2} \right) x^2 y + \left(\frac{c_1(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{e_1}{1 + \lambda_2} \right) x \bar{h} \\
 &\quad + \left(\frac{c_2(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{e_2}{1 + \lambda_2} \right) y \bar{h} + \left(\frac{c_3(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{e_3}{1 + \lambda_2} \right) x^2 \bar{h} \\
 &\quad + \left(\frac{c_4(1 + a_{11})}{a_{12}(1 + \lambda_2)} + \frac{e_4}{1 + \lambda_2} \right) xy \bar{h} + O\left((|u| + |v| + |\bar{h}|)^4\right), \\
 x &= a_{12}(u + v), \quad y = -(1 + a_{11})u + (\lambda_2 - a_{11})v.
 \end{aligned}$$

In order to implement the center manifold theorem, we assume that $W^c(0, 0, 0)$ be the center manifold of (3.11) evaluated at $(0, 0)$ in a small neighborhood of $\bar{h} = 0$, then $W^c(0, 0, 0)$ can be approximated as follows:

$$W^c(0, 0, 0) = \left\{ (u, v, \bar{h}) \in \mathbb{R}^3 : v = m_1 u^2 + m_2 u \bar{h} + m_3 \bar{h}^2 + O\left((|u| + |\bar{h}|)^3\right) \right\},$$

where

$$\begin{aligned}
 m_1 &= a_{12} \left(\frac{a_{13}(1 + a_{11}) + a_{12}a_{23}}{1 - \lambda_2^2} \right) - (1 + a_{11}) \left(\frac{a_{14}(1 + a_{11}) + a_{12}a_{24}}{1 - \lambda_2^2} \right), \\
 m_2 &= \left(\frac{c_1(1 + a_{11}) + e_1 a_{12}}{1 - \lambda_2^2} \right) - (1 + a_{11}) \left(\frac{c_2(1 + a_{11}) + e_2 a_{12}}{a_{12}(1 - \lambda_2^2)} \right), \quad m_3 = 0.
 \end{aligned}$$

Hence, the map restricted to the center manifold $W^c(0, 0, 0)$ is given by

$$F : u \rightarrow -u + k_1 u^2 + k_2 u \bar{h} + k_3 u^2 \bar{h} + k_4 u \bar{h}^2 + k_5 u^3 + O\left((|u| + |\bar{h}|)^4\right),$$

where

$$\begin{aligned}
 k_1 &= a_{12} \left(\frac{a_{13}(\lambda_2 - a_{11}) - a_{12}a_{23}}{1 + \lambda_2} \right) - (1 + a_{11}) \left(\frac{a_{14}(\lambda_2 - a_{11}) - a_{12}a_{24}}{1 + \lambda_2} \right), \\
 k_2 &= \left(\frac{c_1(\lambda_2 - a_{11}) - e_1a_{12}}{1 + \lambda_2} \right) - (1 + a_{11}) \left(\frac{c_2(\lambda_2 - a_{11}) - e_2a_{12}}{a_{12}(1 + \lambda_2)} \right), \\
 k_3 &= 2a_{12}m_2 \left(\frac{a_{13}(\lambda_2 - a_{11}) - a_{12}a_{23}}{1 + \lambda_2} \right) + a_{12} \left(\frac{c_3(\lambda_2 - a_{11}) - e_3a_{12}}{1 + \lambda_2} \right) \\
 &\quad + m_2(\lambda_2 - a_{11}) \left(\frac{a_{14}(\lambda_2 - a_{11}) - a_{12}a_{24}}{1 + \lambda_2} \right) \\
 &\quad - m_2(1 + a_{11}) \left(\frac{a_{14}(\lambda_2 - a_{11}) - a_{12}a_{24}}{1 + \lambda_2} \right) \\
 &\quad - (1 + a_{11}) \left(\frac{c_4(\lambda_2 - a_{11}) - e_4a_{12}}{1 + \lambda_2} \right) + m_1 \left(\frac{c_1(\lambda_2 - a_{11}) - e_1a_{12}}{1 + \lambda_2} \right) \\
 &\quad + m_1(\lambda_2 - a_{11}) \left(\frac{c_2(\lambda_2 - a_{11}) - e_2a_{12}}{a_{12}(1 + \lambda_2)} \right), \\
 k_4 &= m_2 \left(\frac{c_1(\lambda_2 - a_{11}) - e_1a_{12}}{1 + \lambda_2} \right) + m_2(\lambda_2 - a_{11}) \left(\frac{c_2(\lambda_2 - a_{11}) - e_2a_{12}}{a_{12}(1 + \lambda_2)} \right), \\
 k_5 &= 0.
 \end{aligned}$$

Next, we define the following two nonzero real numbers:

$$\begin{aligned}
 l_1 &= \left(\frac{\partial^2 g_1}{\partial u \partial \bar{h}} + \frac{1}{2} \frac{\partial F}{\partial \bar{h}} \frac{\partial^2 F}{\partial u^2} \right)_{(0,0)} = \left(\frac{c_1(\lambda_2 - a_{11}) - e_1a_{12}}{1 + \lambda_2} \right) \\
 &\quad - (1 + a_{11}) \left(\frac{c_2(\lambda_2 - a_{11}) - e_2a_{12}}{a_2(1 + \lambda_2)} \right) \neq 0, \\
 l_2 &= \left(\frac{1}{6} \frac{\partial^3 F}{\partial u^3} + \left(\frac{1}{2} \frac{\partial^2 F}{\partial u^2} \right)^2 \right)_{(0,0)} = k_1^2 > 0.
 \end{aligned}$$

Due to aforementioned computation, we have the following result about period-doubling bifurcation of system (1.2).

Theorem 3.1 *If $l_2 \neq 0$, then system (1.2) undergoes period-doubling bifurcation at the unique positive equilibrium point when parameter h varies in small neighborhood of h_1 . Moreover, the period-two orbits that bifurcate from positive equilibrium are stable.*

4 Neimark–Sacker bifurcation

In this section, we investigate the existence and direction of Neimark–Sacker bifurcation for unique positive steady-state $\left(\beta, \frac{\beta}{k}\right)$ of system (1.2). In order to obtain

the parametric conditions for existence of Neimark–Sacker bifurcation, $P = \left(\beta, \frac{\beta}{k}\right)$ must be non-hyperbolic point such that the Jacobian matrix evaluated at P has pair of complex conjugate eigenvalues with modulus one. Similar type of investigation for discrete-time population models can be found in [15–29].

From (3.1) it is easy to see that $\mathbb{P}(\lambda) = 0$ has two complex conjugate roots with modulus one, if the following conditions are satisfied:

$$\left(k + k^2 - 2\alpha\right)^2 - 8k^2\beta < 0, \quad h = \frac{k^2 + 2\alpha - k}{k(2(\alpha + \beta) - k)} \quad (4.1)$$

Assume that

$$\Omega_{NS} = \left\{ (\alpha, \beta, h_2) \in \mathbb{R}_+^3 : h_2 = \frac{k^2 + 2\alpha - k}{k(2(\alpha + \beta) - k)}, \left(k + k^2 - 2\alpha\right)^2 - 8k^2\beta < 0 \right\}.$$

Due to variation of the parameters in the small neighborhood of the set Ω_{NS} yields Neimark–Sacker bifurcation for positive equilibrium point of system (1.2). Set $h_2 = \frac{k^2 + 2\alpha - k}{k(2(\alpha + \beta) - k)}$ and assume that $(\alpha, \beta, h_2) \in \Omega_{NS}$, then system (1.2) can be expressed by the following two-dimensional map:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \rightarrow \begin{pmatrix} X + h_2(\alpha Y - X + X^2 Y) \\ Y + h_2(\beta - \alpha Y - X^2 Y) \end{pmatrix}. \quad (4.2)$$

Let $\tilde{h}$ denotes the bifurcation parameter, then corresponding perturbed mapping of (4.2) is given as follows:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \rightarrow \begin{pmatrix} X + (h_2 + \tilde{h})(\alpha Y - X + X^2 Y) \\ Y + (h_2 + \tilde{h})(\beta - \alpha Y - X^2 Y) \end{pmatrix}, \quad (4.3)$$

where $|\tilde{h}| \ll 1$ denotes the small perturbation parameter. Next, we consider the transformations $x = X - \beta$ and $y = Y - \frac{\beta}{k}$, then from the map (4.3) we have

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix}, \quad (4.4)$$

where

$$\begin{aligned} f_1(x, y) &= m_{13}x^2 + m_{14}xy + m_{15}x^2y + O((|x| + |y|)^4), \\ f_2(x, y) &= m_{23}x^2 + m_{24}xy + m_{25}x^2y + O((|x| + |y|)^4), \\ m_{11} &= 1 + (h_2 + \tilde{h})\left(\frac{2\beta^2}{k} - 1\right), \quad m_{12} = (h_2 + \tilde{h})k, \quad m_{13} = \frac{(h_2 + \tilde{h})\beta}{k}, \\ m_{14} &= 2(h_2 + \tilde{h})\beta, \quad m_{15} = h_2 + \tilde{h}, \quad m_{21} = -\frac{2(h_2 + \tilde{h})\beta^2}{k}, \quad m_{22} = 1 - (h_2 + \tilde{h})k, \\ m_{23} &= -\frac{(h_2 + \tilde{h})\beta}{k}, \quad m_{24} = -2(h_2 + \tilde{h})\beta, \quad m_{25} = -(h_2 + \tilde{h}). \end{aligned}$$

The characteristic equation of Jacobian matrix of linearized system of (4.4) evaluated at the equilibrium (0, 0) can be written as follows:

$$\lambda^2 - A(\tilde{h})\lambda + B(\tilde{h}) = 0, \quad (4.5)$$

where

$$A(\tilde{h}) = 2 - (h_2 + \tilde{h}) \left(k - 1 + \frac{2\alpha}{k} \right),$$

and

$$B(\tilde{h}) = \frac{(1 - (h_2 + \tilde{h})k)(k + (h_2 + \tilde{h})k - 2(h_2 + \tilde{h})\alpha)}{k} + 2(h_2 + \tilde{h})^2\beta.$$

Since $(\alpha, \beta, h_2) \in \Omega_{NS}$, therefore the complex conjugate roots of (4.5) are given by:

$$\lambda_1 = \frac{A(\tilde{h}) - i\sqrt{4B(\tilde{h}) - (A(\tilde{h}))^2}}{2},$$

and

$$\lambda_2 = \frac{A(\tilde{h}) + i\sqrt{4B(\tilde{h}) - (A(\tilde{h}))^2}}{2}.$$

Then, we obtain that

$$|\lambda_1| = |\lambda_2| = \sqrt{\frac{(1 - (h_2 + \tilde{h})k)(k + (h_2 + \tilde{h})k - 2(h_2 + \tilde{h})\alpha)}{k} + 2(h_2 + \tilde{h})^2\beta}.$$

Moreover, in order to obtain the nondegeneracy conditions, first we have

$$\left(\frac{d|\lambda_2|}{d\tilde{h}} \right)_{\tilde{h}=0} = \left(\frac{d|\lambda_1|}{d\tilde{h}} \right)_{\tilde{h}=0} = \frac{1}{2} \left(k - 1 + \frac{2\alpha}{k} \right) \neq 0.$$

Moreover, we have $-2 < A(0) < 2$ because $(\alpha, \beta, h_2) \in \Omega_{NS}$. On the other hand, we have $A(0) = \frac{k^4 - 4k\alpha + 4\alpha^2 + k^2(1 - 4\beta)}{k^2(k - 2(\alpha + \beta))}$. Assume that $A(0) \neq 0$ and $A(0) \neq -1$, that is,

$$k^4 - 4k\alpha + 4\alpha^2 + k^2(1 - 4\beta) \neq 0, \quad \frac{k^4 - 4k\alpha + 4\alpha^2 + k^2(1 - 4\beta)}{k^2(k - 2(\alpha + \beta))} \neq -1. \quad (4.6)$$

Conditions (4.6) together $(\alpha, \beta, h_2) \in \Omega_{NS}$ make sure that $A(0) \neq \pm 2, 0, -1$, and in a result we have $\lambda_1^m, \lambda_2^m \neq 1$ for all $m = 1, 2, 3, 4$ at $\tilde{h} = 0$. Hence roots of (4.5)

do not lie in the intersection of the unit circle with the coordinate axes when $\tilde{h} = 0$. In order to obtain the normal form of (4.4) at $\tilde{h} = 0$, assuming that $\kappa = \frac{A(0)}{2}$ and $\omega = \frac{\sqrt{4B(0)-(A(0))^2}}{2}$. Furthermore, considering the following transformation:

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} m_{12} & 0 \\ \kappa - m_{11} & -\omega \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (4.7)$$

Under transformation (4.7), the normal form of (4.4) can be written as:

$$\begin{pmatrix} u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} \kappa & -\omega \\ \omega & \kappa \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} \tilde{f}(u, v) \\ \tilde{g}(u, v) \end{pmatrix}, \quad (4.8)$$

where

$$\begin{aligned} \tilde{f}(u, v) &= \frac{m_{13}}{m_{12}}x^2 + \frac{m_{14}}{m_{12}}xy + \frac{m_{15}}{m_{12}}x^2y + O((|u| + |v|)^4), \\ \tilde{g}(u, v) &= \left(\frac{(\kappa - m_{11})m_{13}}{m_{12}\omega} - \frac{m_{23}}{\omega} \right)x^2 + \left(\frac{(\kappa - m_{11})m_{14}}{m_{12}\omega} - \frac{m_{24}}{\omega} \right)xy \\ &\quad + \left(\frac{(\kappa - m_{11})m_{15}}{m_{12}\omega} - \frac{m_{25}}{\omega} \right)x^2y + O((|u| + |v|)^4), \end{aligned}$$

$x = m_{12}u$ and $y = (\kappa - m_{11})u - \omega v$. Next, we consider the following real number:

$$L = \left(\left[-\operatorname{Re} \left(\frac{(1 - 2\lambda_1)\lambda_2^2}{1 - \lambda_1} \zeta_{20}\zeta_{11} \right) - \frac{1}{2}|\zeta_{11}|^2 - |\zeta_{02}|^2 + \operatorname{Re}(\lambda_2\zeta_{21}) \right] \right)_{\tilde{h}=0},$$

where

$$\begin{aligned} \zeta_{20} &= \frac{1}{8} [\tilde{f}_{uu} - \tilde{f}_{vv} + 2\tilde{g}_{uv} + i(\tilde{g}_{uu} - \tilde{g}_{vv} - 2\tilde{f}_{uv})], \\ \zeta_{11} &= \frac{1}{4} [\tilde{f}_{uu} + \tilde{f}_{vv} + i(\tilde{g}_{uu} + \tilde{g}_{vv})], \\ \zeta_{02} &= \frac{1}{8} [\tilde{f}_{uu} - \tilde{f}_{vv} - 2\tilde{g}_{uv} + i(\tilde{g}_{uu} - \tilde{g}_{vv} + 2\tilde{f}_{uv})], \\ \zeta_{21} &= \frac{1}{16} [\tilde{f}_{uuu} + \tilde{f}_{uvv} + \tilde{g}_{uuv} + \tilde{g}_{vvv} + i(\tilde{g}_{uuu} + \tilde{g}_{uvv} - \tilde{f}_{uuv} - \tilde{f}_{vvv})]. \end{aligned}$$

Moreover, the aforementioned partial derivatives of $\tilde{f}$ and $\tilde{g}$ at $\tilde{h} = 0$ are given by:

$$\begin{aligned}\tilde{f}_{uu} &= \frac{2h_2\beta(k(3h_2 - 2 + 2\omega)) - 4h_2\beta^2}{k}, \\ \tilde{g}_{uv} &= -\frac{2h_2\beta(h_2k^2 - 2h_2\beta^2 + k(h_2 - 1 + \kappa))}{k}, \\ \tilde{f}_{vv} &= 0, \quad \tilde{g}_{uu} = \frac{2h_2\beta(h_2k^2 - 2h_2\beta^2 + k(h_2 - 1 + \kappa))(k(3h_2 - 2 + 2\kappa) - 4h_2\beta^2)}{k^2\omega}, \\ \tilde{g}_{vv} &= 0, \quad \tilde{f}_{uv} = -2h_2\beta\kappa, \quad \tilde{f}_{uuu} = 6h_2^2(k(h_2 - 1 + \kappa) - 2h_2\beta^2), \\ \tilde{f}_{uvv} &= 0, \quad \tilde{g}_{uuv} = -2h_2^2(h_2k^2 - 2h_2\beta^2 + k(h_2 - 1 + \kappa)), \\ \tilde{g}_{uuu} &= \frac{6h_2^2(k(h_2 - 1 + \kappa) - 2h_2\beta^2)(h_2k^2 - 2h_2\beta^2 + k(h_2 - 1 + \kappa))}{k\omega}, \\ \tilde{g}_{vvv} &= \tilde{g}_{uvv} = \tilde{f}_{vvv} = 0, \quad \tilde{f}_{uuv} = -2h_2^2k\kappa.\end{aligned}$$

Arguing as in [30–34], we have the following result which gives parametric conditions for existence and direction of Neimark–Sacker bifurcation for positive equilibrium of system (1.2).

Theorem 4.1 Suppose that (4.6) holds true and $L \neq 0$, then system (1.2) endures Neimark–Sacker bifurcation at its unique positive steady-state when the bifurcation parameter h varies in a small neighborhood of $h_2 = \frac{k^2 + 2\alpha - k}{k(2(\alpha + \beta) - k)}$. Furthermore, if $L < 0$, then an attracting invariant closed curve bifurcates from the equilibrium point for $h > h_2$, and if $L > 0$, then a repelling invariant closed curve bifurcates from the equilibrium point for $h < h_2$.

At the end of this section, we investigate the parametric conditions under which the system (1.3) undergoes Neimark–Sacker bifurcation at its positive steady-state. For this, we take h as bifurcation parameter and assume that:

$$(\Delta(\alpha, \beta, h))^2 < 4 \left(\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} \right). \quad (4.9)$$

Suppose that (4.9) holds true, then characteristic polynomial (2.6) has pair of complex roots. Furthermore, we suppose that the following condition holds true:

$$\frac{\alpha + (1 + 3h)\beta^2}{(\alpha + h\alpha + \beta^2 + 2h\beta^2)(1 + h(\alpha + \beta^2))} = 1. \quad (4.10)$$

Combining (4.9) and (4.10), we define the following set:

$$\Upsilon_{NS} = \left\{ (\alpha, \beta, h) : \Delta(\alpha, \beta, h) < 2, \quad h = \frac{\beta^2 - \alpha - \alpha^2 - 2\alpha\beta^2 - \beta^4}{\alpha^2 + 3\alpha\beta^2 + 2\beta^4}, \quad \alpha, \beta, h > 0 \right\}.$$

Suppose that $(\alpha, \beta, h) \in \Upsilon_{NS}$, then system (1.3) undergoes Neimark–Sacker bifurcation as the bifurcation parameter h varies in a small neighborhood of h_3 given by:

$$h_3 = \frac{\beta^2 - \alpha - \alpha^2 - 2\alpha\beta^2 - \beta^4}{\alpha^2 + 3\alpha\beta^2 + 2\beta^4}.$$

5 Chaos control

Controlling chaos and bifurcation is considered an important feature for population models particularly when these models are related to biological breeding of species. Usually discrete-time models have more complex behavior as compare to continuous one. In order to avoid the population from unpredictable situations, it is necessary to implement the chaos control techniques. In this section, we study two feedback control strategies in order to move the unstable trajectory towards the stable one. For this, first we apply the OGY method to system (1.2). This method was proposed by Ott et al. [35]. Also see [36] for further details related to OGY method. In order to apply OGY method to system (1.2), we rewrite system (1.2) as follows:

$$\begin{aligned} x_{n+1} &= x_n + h \left(\alpha y_n - x_n + x_n^2 y_n \right) = f(x_n, y_n, \alpha), \\ y_{n+1} &= y_n + h \left(\beta - \alpha y_n - x_n^2 y_n \right) = g(x_n, y_n, \alpha), \end{aligned} \quad (5.1)$$

where α is taken as controlling parameter through which one can achieve desired chaos control by means of applying only very small perturbations. For this, we restrict α to lie in some small interval $\alpha \in \left(\alpha_0 - \eta, \alpha_0 + \eta \right)$ with $\eta > 0$ and α_0 denotes the nominal value belong to chaotic region. We apply the stabilizing feedback control strategy in order to move the trajectory towards the desired orbit. Assume that $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ be unstable equilibrium point of system (1.2) in chaotic region produced by the emergence of Neimark–Sacker bifurcation, then system (5.1) can be approximated in the neighborhood of the unstable equilibrium point $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ by the following linear map:

$$\begin{bmatrix} x_{n+1} - x^* \\ y_{n+1} - y^* \end{bmatrix} \approx J(x^*, y^*, \alpha_0) \begin{bmatrix} x_n - x^* \\ y_n - y^* \end{bmatrix} + B[\alpha - \alpha_0], \quad (5.2)$$

where

$$J(x^*, y^*, \alpha_0) = \begin{bmatrix} \frac{\partial f(x^*, y^*, \alpha_0)}{\partial x} & \frac{\partial f(x^*, y^*, \alpha_0)}{\partial y} \\ \frac{\partial g(x^*, y^*, \alpha_0)}{\partial x} & \frac{\partial g(x^*, y^*, \alpha_0)}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 - h + \frac{2h\beta^2}{\alpha_0 + \beta^2} & h(\alpha_0 + \beta^2) \\ -\frac{2h\beta^2}{\alpha_0 + \beta^2} & 1 - h(\alpha_0 + \beta^2) \end{bmatrix}$$

and

$$B = \begin{bmatrix} \frac{\partial f(x^*, y^*, \alpha_0)}{\partial \alpha} \\ \frac{\partial g(x^*, y^*, \alpha_0)}{\partial \alpha} \end{bmatrix} = \begin{bmatrix} \frac{h\beta}{\alpha_0 + \beta^2} \\ -\frac{h\beta}{\alpha_0 + \beta^2} \end{bmatrix}.$$

Moreover, system (5.1) is controllable provided that the following matrix

$$C = [B : JB] = \begin{bmatrix} \frac{h\beta}{\alpha_0 + \beta^2} & h\beta \left(-h + \frac{\alpha_0 - h\alpha_0 + (1+h)\beta^2}{(\alpha_0 + \beta^2)^2} \right) \\ -\frac{h\beta}{\alpha_0 + \beta^2} & \frac{h\beta(-2h\beta^2 + (\alpha_0 + \beta^2)(-1 + h(\alpha_0 + \beta^2)))}{(\alpha_0 + \beta^2)^2} \end{bmatrix} \quad (5.3)$$

is of rank 2. Moreover, it must be noted that h cannot be taken as control parameter in this case in order to implement the OGY method to system (1.2). Indeed, if we take $f(x_n, y_n, h) = x_n + h(\alpha y_n - x_n + x_n^2 y_n)$ and $g(x_n, y_n, h) = y_n + h(\beta - \alpha y_n - x_n^2 y_n)$, then rank of controllability matrix C reduces to zero. So, α is taken as control parameter in order to apply OGY method to system (1.2).

Moreover, taking $[\alpha - \alpha_0] = -K \begin{bmatrix} x_n - x^* \\ y_n - y^* \end{bmatrix}$, where $K = [\rho_1 \ \rho_2]$, then system (5.5) can be written as

$$\begin{bmatrix} x_{n+1} - x^* \\ y_{n+1} - y^* \end{bmatrix} \approx [J - BK] \begin{bmatrix} x_n - x^* \\ y_n - y^* \end{bmatrix}. \quad (5.4)$$

Furthermore, the corresponding controlled system of (1.2) is given by

$$\begin{aligned} x_{n+1} &= x_n + h \left((\alpha_0 - \rho_1(x_n - x^*) - \rho_2(y_n - y^*))y_n - x_n + x_n^2 y_n \right), \\ y_{n+1} &= y_n + h \left(\beta - (\alpha_0 - \rho_1(x_n - x^*) - \rho_2(y_n - y^*))y_n - x_n^2 y_n \right). \end{aligned} \quad (5.5)$$

Moreover, equilibrium point $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ is locally asymptotically stable if and only if both eigenvalues of the matrix $J - BK$ lie in an open unit disk. The Jacobian matrix $J - BK$ of the controlled system (5.5) can be written as follows:

$$J - BK = \begin{bmatrix} 1 - h + \frac{2h\beta^2}{\alpha_0 + \beta^2} - \frac{h\beta\rho_1}{\alpha_0 + \beta^2} & h(\alpha_0 + \beta^2) - \frac{h\beta\rho_2}{\alpha_0 + \beta^2} \\ -\frac{2h\beta^2}{\alpha_0 + \beta^2} + \frac{h\beta\rho_1}{\alpha_0 + \beta^2} & 1 - h(\alpha_0 + \beta^2) + \frac{h\beta\rho_2}{\alpha_0 + \beta^2} \end{bmatrix}.$$

The characteristic equation of the Jacobian matrix $J - BK$ is given by

$$\begin{aligned} \mathbb{P}(\lambda) &= \lambda^2 - \left(2 - h - h\alpha_0 - h\beta^2 + \frac{2h\beta^2}{\alpha_0 + \beta^2} - \frac{h\beta\rho_1}{\alpha_0 + \beta^2} + \frac{h\beta\rho_2}{\alpha_0 + \beta^2} \right) \lambda \\ &\quad + 1 - h - h\alpha_0 + h^2\alpha_0 - h\beta^2 + h^2\beta^2 \\ &\quad + \frac{2h\beta^2}{\alpha_0 + \beta^2} - \frac{h\beta\rho_1}{\alpha_0 + \beta^2} + \frac{h\beta\rho_2}{\alpha_0 + \beta^2} - \frac{h^2\beta\rho_2}{\alpha_0 + \beta^2} = 0. \end{aligned} \quad (5.6)$$

Let λ_1 and λ_2 are the roots of characteristic equation (5.6), then we have

$$\lambda_1 + \lambda_2 = 2 - h - h\alpha_0 - h\beta^2 + \frac{2h\beta^2}{\alpha_0 + \beta^2} - \frac{h\beta\rho_1}{\alpha_0 + \beta^2} + \frac{h\beta\rho_2}{\alpha_0 + \beta^2}, \quad (5.7)$$

and

$$\begin{aligned} \lambda_1 \lambda_2 = 1 - h - h\alpha_0 + h^2\alpha_0 - h\beta^2 + h^2\beta^2 + \frac{2h\beta^2}{\alpha_0 + \beta^2} - \frac{h\beta\rho_1}{\alpha_0 + \beta^2} \\ + \frac{h\beta\rho_2}{\alpha_0 + \beta^2} - \frac{h^2\beta\rho_2}{\alpha_0 + \beta^2}. \end{aligned} \quad (5.8)$$

Next we take $\lambda_1 = \pm 1$ and $\lambda_1 \lambda_2 = 1$ in order to obtain the lines of marginal stability for corresponding controlled system. Moreover, these restrictions make sure that λ_1 and λ_2 inside the open unit disk. Assuming that $\lambda_1 \lambda_2 = 1$, then Eq. (5.8) implies that:

$$L_1: h\alpha_0 + h\beta^2 + \frac{2\beta^2}{\alpha_0 + \beta^2} + \frac{\beta\rho_2}{\alpha_0 + \beta^2} = 1 + \alpha_0 + \beta^2 + \frac{h\beta\rho_2}{\alpha_0 + \beta^2} + \frac{\beta\rho_1}{\alpha_0 + \beta^2}.$$

Moreover, we assume that $\lambda_1 = 1$, then (5.7) and (5.8) yield that:

$$L_2: \alpha_0 + \beta^2 = \frac{\beta\rho_2}{\alpha_0 + \beta^2}.$$

Finally taking $\lambda_1 = -1$, then from Eqs. (5.7) and (5.8) we get

$$\begin{aligned} L_3: 4 - 2h - 2h\alpha_0 + h^2\alpha_0 - 2h\beta^2 + h^2\beta^2 + \frac{4h\beta^2}{\alpha_0 + \beta^2} - \frac{2h\beta\rho_1}{\alpha_0 + \beta^2} \\ + \frac{2h\beta\rho_2}{\alpha_0 + \beta^2} - \frac{h^2\beta\rho_2}{\alpha_0 + \beta^2} = 0. \end{aligned}$$

Then, stable eigenvalues lie within the triangular region in $\rho_1 \rho_2$ -plane bounded by the straight lines L_1, L_2, L_3 for particular parametric values.

Secondly, in order to control the chaos due to emergence of bifurcation in system (1.2), we apply hybrid control feedback methodology [37]. This strategy was basically formulated for controlling the period-doubling bifurcation. Moreover, in [29] same technique is implemented for controlling the chaos due to emergence of Neimark–Sacker bifurcation. Assume that system (1.2) undergoes bifurcation at equilibrium point $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2}\right)$, then corresponding controlled system can be written as:

$$\begin{aligned} x_{n+1} &= \rho \left[x_n + h \left(\alpha y_n - x_n + x_n^2 y_n \right) \right] + (1 - \rho)x_n, \\ y_{n+1} &= \rho \left[y_n + h \left(\beta - \alpha y_n - x_n^2 y_n \right) \right] + (1 - \rho)y_n, \end{aligned} \quad (5.9)$$

where $0 < \rho < 1$ and controlled strategy in (5.9) is a combination of both parameter perturbation and feedback control. Moreover, by suitable choice of controlled parameter ρ , the period-doubling bifurcation and Neimark–Sacker bifurcation for the equilibrium point (x^*, y^*) of controlled system (5.9) can be advanced (delayed) or

even completely eliminated. The Jacobian matrix of controlled system (5.9) evaluated at positive equilibrium (x^*, y^*) is given by

$$\begin{bmatrix} 1 + \frac{h(\beta^2 - \alpha)\rho}{\alpha + \beta^2} & h(\alpha + \beta^2)\rho \\ -\frac{2h\beta^2\rho}{\alpha + \beta^2} & 1 - h(\alpha + \beta^2)\rho \end{bmatrix}. \quad (5.10)$$

Moreover, the characteristic equation of Jacobian matrix of the controlled system (5.9) is given by

$$\begin{aligned} \lambda^2 - \left(2 - h\alpha\rho - h\beta^2\rho - \frac{h\alpha\rho}{\alpha + \beta^2} + \frac{h\beta^2\rho}{\alpha + \beta^2} \right) \lambda + 1 - h\alpha\rho - h\beta^2\rho \\ - \frac{h\alpha\rho}{\alpha + \beta^2} + \frac{h\beta^2\rho}{\alpha + \beta^2} + 2h^2\beta^2\rho^2 + \frac{h^2\alpha^2\rho^2}{\alpha + \beta^2} - \frac{h^2\beta^4\rho^2}{\alpha + \beta^2} = 0. \end{aligned}$$

The following result gives conditions for local asymptotic stability of positive equilibrium $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ of the controlled system (5.9).

Lemma 5.1 *The positive equilibrium $(x^*, y^*) = \left(\beta, \frac{\beta}{\alpha + \beta^2} \right)$ of the controlled system (5.9) is locally asymptotically stable, if the following condition holds:*

$$\begin{aligned} \left| 2 - h\alpha\rho - h\beta^2\rho - \frac{h\alpha\rho}{\alpha + \beta^2} + \frac{h\beta^2\rho}{\alpha + \beta^2} \right| < 2 - h\alpha\rho - h\beta^2\rho - \frac{h\alpha\rho}{\alpha + \beta^2} \\ + \frac{h\beta^2\rho}{\alpha + \beta^2} + 2h^2\beta^2\rho^2 + \frac{h^2\alpha^2\rho^2}{\alpha + \beta^2} - \frac{h^2\beta^4\rho^2}{\alpha + \beta^2} < 2. \end{aligned}$$

6 Numerical simulation and discussion

Example 6.1 Let $\alpha = 2.05$, $\beta = 1.61$, $h \in [0, 1]$ and $(x_0, y_0) = (1.61, 0.37)$, then system (1.2) endures period-bifurcation bifurcation as $h \approx 0.6771173643330266$. The corresponding bifurcation diagrams and maximum Lyapunov exponents (MLE) are shown in Fig. 1. Moreover, at $(\alpha, \beta, h) = (2.05, 1.61, 0.6771173643330266)$ the system (1.2) has unique positive equilibrium point $(1.61, 0.346826)$ and characteristic equation of the Jacobian matrix of the system (1.2) evaluated at this positive equilibrium is given by

$$\lambda^2 + 1.0641733985099622\lambda + 0.06417339850996301 = 0. \quad (6.1)$$

Furthermore, the roots of (6.1) are $\lambda_1 = -1$ and $\lambda_2 = -0.0641734$ with $|\lambda_2| \neq 1$. Thus, we have $(\alpha, \beta, h) = (2.05, 1.61, 0.6771173643330266) \in \Omega_{FB}$.

Example 6.2 Let $\alpha = 0.2$, $\beta = 0.9$, $h \in [0.1, 0.6]$ and $(x_0, y_0) = (0.9, 0.89)$, then system (1.2) endures Neimark–Sacker bifurcation as $h \approx 0.4020194098617782$. The corresponding bifurcation diagrams and maximum Lyapunov exponents (MLE) are

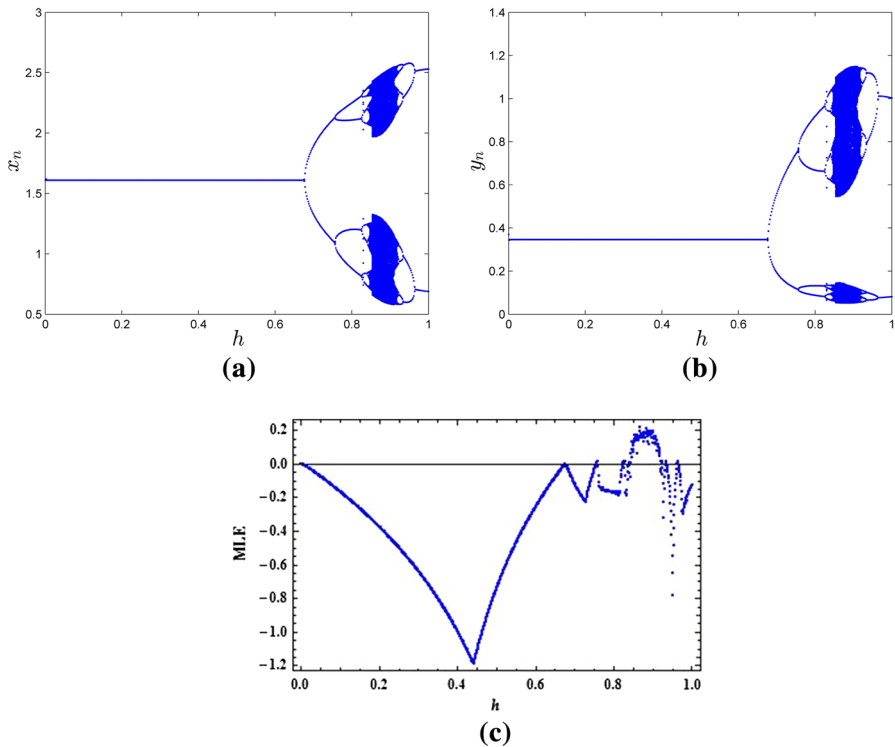


Fig. 1 Bifurcation diagrams and MLE for system (1.2) with $\alpha = 2.05$, $\beta = 1.61$, $h \in [0, 1]$ and $(x_0, y_0) = (1.61, 0.37)$. **a** Bifurcation diagram for x_n . **b** Bifurcation diagram for y_n . **c** Maximum Lyapunov exponents

shown in Fig. 2. Moreover, at $(\alpha, \beta, h) = (0.2, 0.9, 0.4020194098617782)$ the system (1.2) has unique positive equilibrium point $(0.9, 0.891089)$ and characteristic equation of the Jacobian matrix of the system (1.2) evaluated at this positive equilibrium is given by

$$\lambda^2 - 1.8367641980353318\lambda + 1 = 0. \quad (6.2)$$

Furthermore, the roots of (6.2) are $\lambda_1 = 0.9183820990176659 + 0.3956947310792831i$ and $\lambda_2 = 0.9183820990176659 - 0.3956947310792831i$ with $|\lambda_{1,2}| = 1$. Thus, we have $(\alpha, \beta, h) = (0.2, 0.9, 0.4020194098617782) \in \Omega_{NB}$. Moreover, keeping fixed the parametric values $\alpha = 0.2$ and $\beta = 0.9$, while varying values of bifurcation parameter $h \in [0.1, 0.6]$, the phase portraits of system (1.2) are depicted in Fig. 3.

Example 6.3 Next we assume that $h = 0.35$, $\beta = 0.8$, $\alpha \in [0.1, 0.35]$ and $(x_0, y_0) = (0.8, 0.75)$, then system (1.2) undergoes Neimark–Sacker bifurcation. For these parametric values, bifurcation diagrams and corresponding maximum Lyapunov exponents are plotted in Fig. 4. At $(\alpha, \beta, h) = (0.19106506140871923, 0.8, 0.35)$, system (1.2) has unique positive equilibrium point $(0.8, 0.96262)$, and characteristic equation of the Jacobian matrix evaluated at this equilibrium point is given by:

$$\lambda^2 - 1.898194529977432\lambda + 1 = 0. \quad (6.3)$$

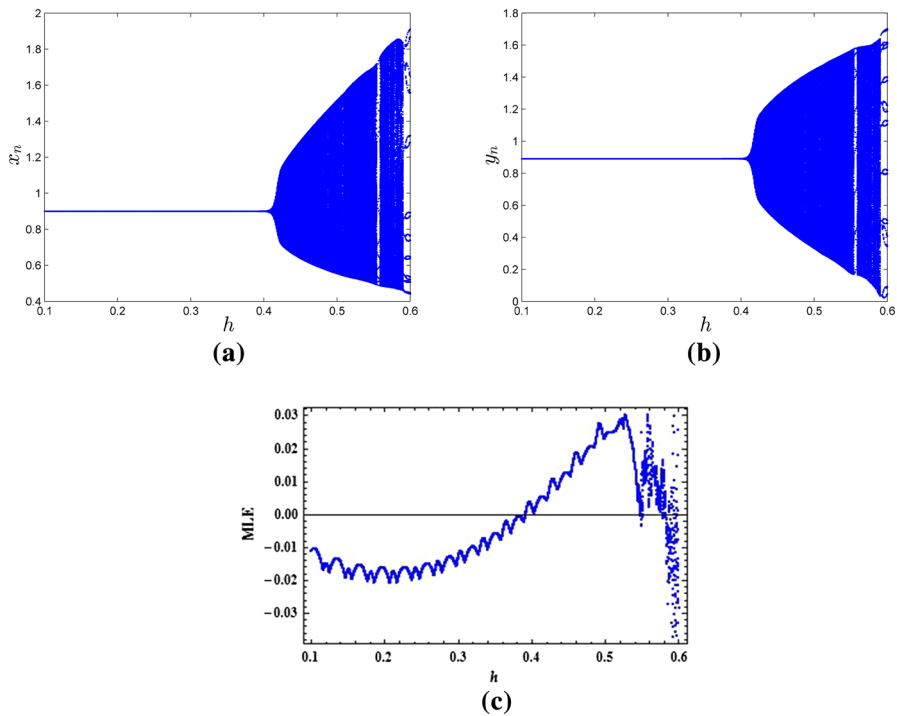


Fig. 2 Bifurcation diagrams and MLE for system (1.2) with $\alpha = 0.2, \beta = 0.9, h \in [0.1, 0.6]$ and $(x_0, y_0) = (0.9, 0.89)$. **a** Bifurcation diagram for x_n . **b** Bifurcation diagram for y_n . **c** Maximum Lyapunov exponents

The roots of (6.3) are $\lambda_1 = 0.9491 - 0.31498i$ and $\lambda_2 = 0.9491 + 0.31498i$ with $|\lambda_{1,2}| = 1$. In order to discuss the OGY feedback control method for system (1.2), we take $\alpha_0 = 0.1$. Then corresponding controlled system is given by:

$$\begin{aligned} x_{n+1} &= x_n + 0.35 \left((0.1 - \rho_1(x_n - 0.8) - \rho_2(y_n - 1.08108))y_n - x_n + x_n^2 y_n \right), \\ y_{n+1} &= y_n + 0.35 \left(0.8 - (0.1 - \rho_1(x_n - 0.8) - \rho_2(y_n - 1.08108))y_n - x_n^2 y_n \right). \end{aligned} \quad (6.4)$$

Then, Jacobian matrix $J - BK$ of the controlled system (6.4) is given by

$$J - BK = \begin{bmatrix} 1.25541 - 0.378378\rho_1 & 0.259 - 0.378378\rho_2 \\ -0.605405 + 0.378378\rho_1 & 0.741 + 0.378378\rho_2 \end{bmatrix}.$$

Moreover, the lines L_1, L_2 and L_3 for marginal stability are given by

$$L_1: 0.245946\rho_2 = 0.378378\rho_1 - 0.0870554,$$

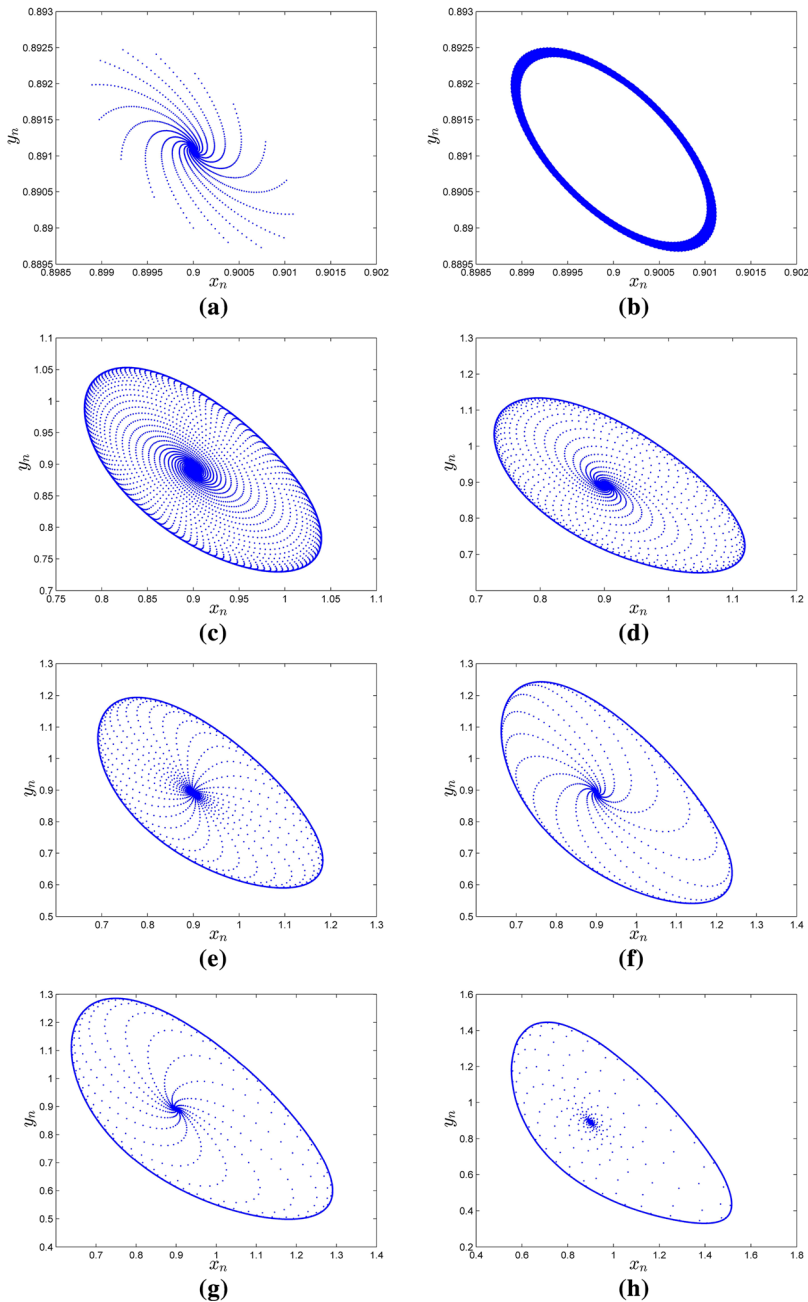


Fig. 3 Phase portraits of system (1.2) for $\alpha = 0.2$, $\beta = 0.9$, $x_0 = 0.9$, $y_0 = 0.89$ and with different values of h . **a** Phase portrait for $h = 0.39$. **b** Phase portrait for $h = 0.4020194098617782$. **c** Phase portrait for $h = 0.41$. **d** Phase portrait for $h = 0.42$. **e** Phase portrait for $h = 0.43$. **f** Phase portrait for $h = 0.44$. **g** Phase portrait for $h = 0.45$. **h** Phase portrait for $h = 0.5$. **i** Phase portrait for $h = 0.58$. **j** Phase portrait for $h = 0.6$

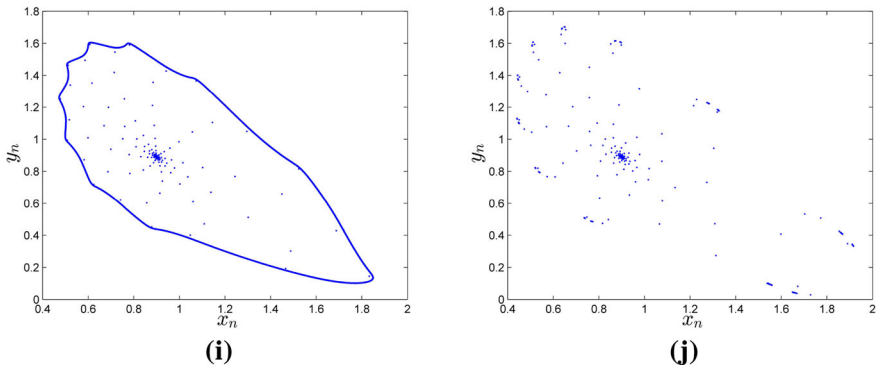


Fig. 3 continued

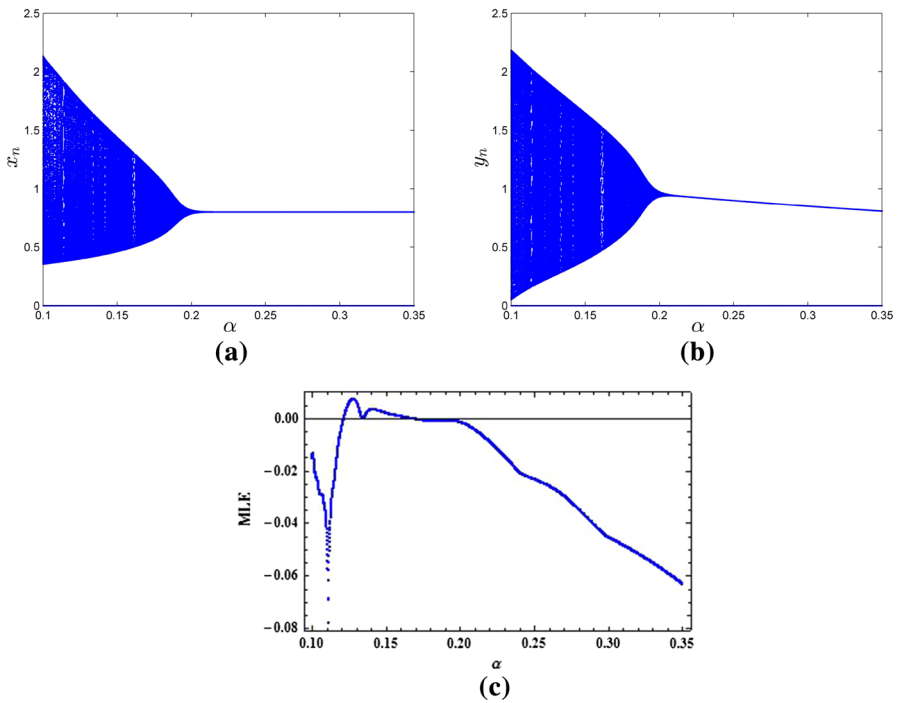


Fig. 4 Bifurcation diagrams and MLE for system (1.2) with $h = 0.35$, $\beta = 0.8$, $\alpha \in [0.1, 0.35]$ and $(x_0, y_0) = (0.8, 0.75)$. **a** Bifurcation diagram for x_n . **b** Bifurcation diagram for y_n . **c** Maximum Lyapunov exponents

$$L_2: 0.132432\rho_2 = 0.09065,$$

and

$$L_3: 0.624324\rho_2 = 0.756757\rho_1 - 4.08346.$$

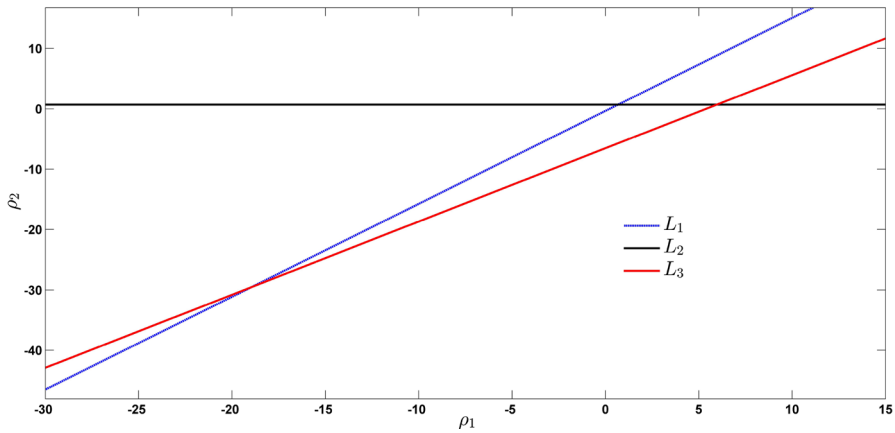


Fig. 5 Triangular stability region bounded by L_1 , L_2 and L_3 for the controlled system (6.4)

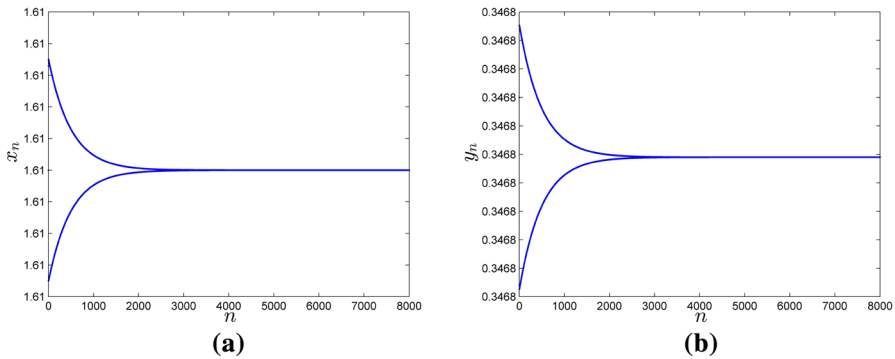


Fig. 6 Plots for the controlled system (6.5) with $\rho = 0.999$ and $(x_0, y_0) = (1.61, 0.346826)$. **a** Plot of x_n for the controlled system (6.5). **b** Plot of y_n for the controlled system (6.5)

The triangular stability region bounded by these marginal lines is shown in Fig. 5.

Next, we again take $\alpha = 2.05$, $\beta = 1.61$, $h = 0.6771173643330266$ and $(x_0, y_0) = (1.61, 0.37)$, then system (1.2) undergoes period-bifurcation bifurcation for these parametric values. In this case, we implement hybrid control strategy to move the unstable equilibrium point $(1.61, 0.346826)$ towards the stable trajectory. The corresponding controlled system is given by

$$\begin{aligned} x_{n+1} &= \rho \left[x_n + 0.6771173643330266 \left(2.05y_n - x_n + x_n^2 y_n \right) \right] + (1 - \rho)x_n, \\ y_{n+1} &= \rho \left[y_n + 0.6771173643330266 \left(1.61 - 2.05y_n - x_n^2 y_n \right) \right] + (1 - \rho)y_n. \end{aligned} \quad (6.5)$$

The Jacobian matrix of the controlled system (6.5) evaluated at equilibrium point $(1.61, 0.346826)$ is given by

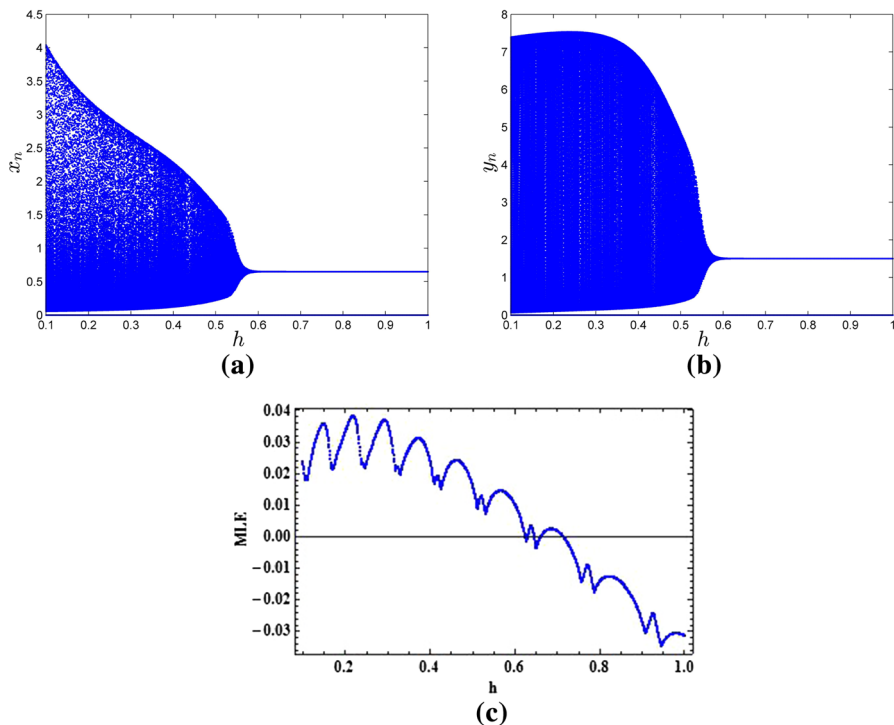


Fig. 7 Bifurcation diagrams and MLE for system (1.3) with $\alpha = 0.01$, $\beta = 0.65$, $h \in [0.1, 1]$ and $(x_0, y_0) = (0.65, 1.5)$. **a** Bifurcation diagram for x_n . **b** Bifurcation diagram for y_n . **c** Maximum Lyapunov exponents

$$M = \begin{bmatrix} 1 + 0.0790731\rho & 3.14325\rho \\ -0.75619\rho & 1 - 3.14325\rho \end{bmatrix}.$$

Then, characteristic equation of Jacobian matrix M is given by

$$\lambda^2 - (2 - 3.06417\rho)\lambda + 1 - 3.06417\rho + 2.12835\rho^2 = 0. \quad (6.6)$$

Furthermore, roots of (6.6) lie inside the unit open disk if $0 < \rho < 1$. Thus period-doubling bifurcation is completely eliminated in allowable interval for control parameter ρ . Let $\rho = 0.999$ and $(x_0, y_0) = (1.61, 0.346826)$, then plots of x_n and y_n are depicted in Fig. 6 for the controlled system (6.5).

Example 6.4 Finally, we consider system (1.3). Assuming $\alpha = 0.01$, $\beta = 0.65$, $h \in [0.1, 1]$ and with initial values $(x_0, y_0) = (0.65, 1.5)$. Then, system (1.3) undergoes Neimark–Sacker bifurcation as the bifurcation parameter h passes through a critical values $h \approx 0.6096575736064629$. Moreover, at $(\alpha, \beta, h) = (0.01, 0.65, 0.6096575736064629)$ system (1.3) has unique positive steady-state which is given by $(0.65, 1.5028901734104045)$. For the parametric values $(\alpha, \beta, h) =$

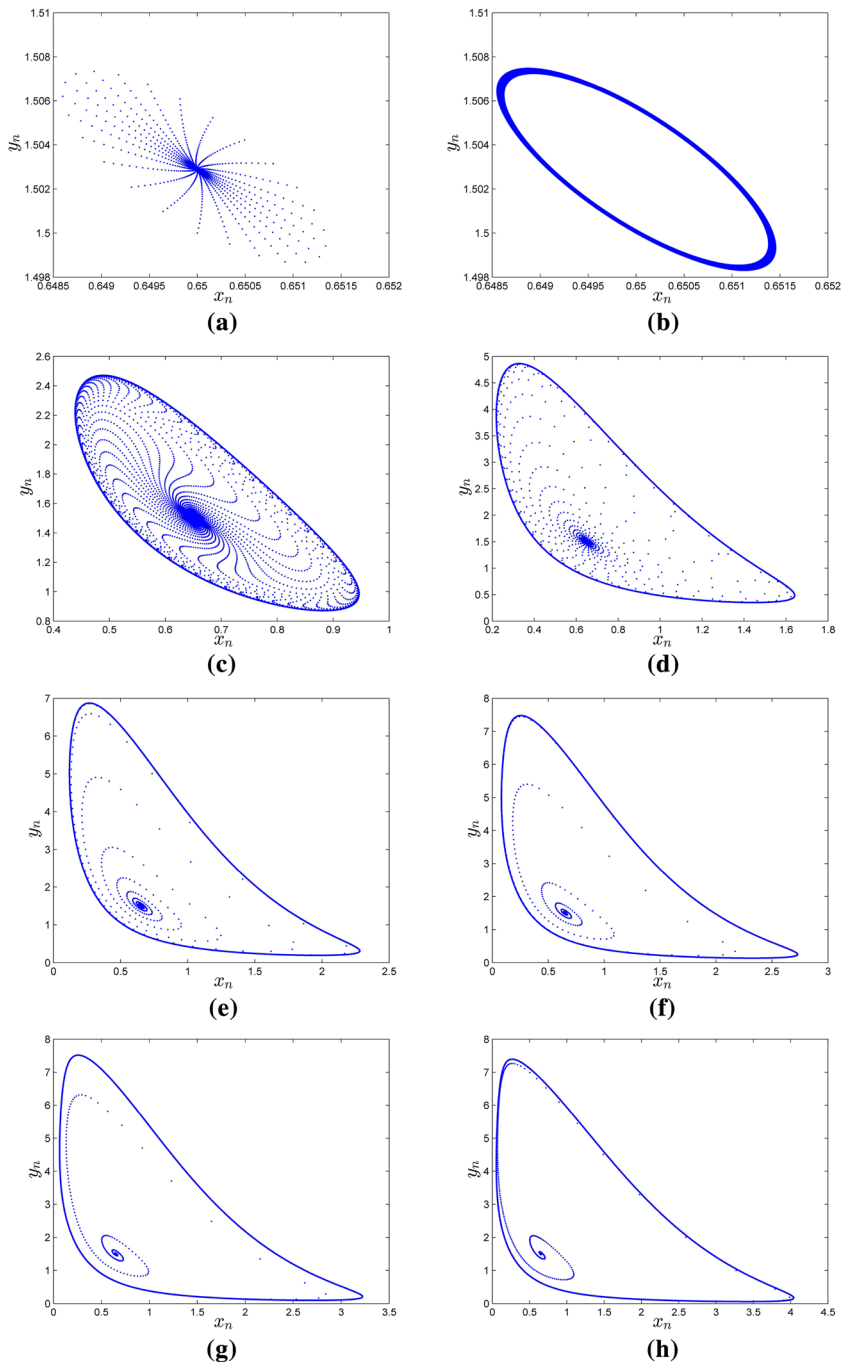


Fig. 8 Phase portraits of (1.3) for $\alpha = 0.01$, $\beta = 0.65$, $(x_0, y_0) = (0.65, 1.5)$ and with different values of h . **a** Phase portrait for $h = 0.65$. **b** Phase portrait for $h = 0.6096575736064629$. **c** Phase portrait for $h = 0.59$. **d** Phase portrait for $h = 0.5$. **e** Phase portrait for $h = 0.4$. **f** Phase portrait for $h = 0.3$. **g** Phase portrait for $h = 0.2$. **h** Phase portrait for $h = 0.1$

(0.01, 0.65, 0.6096575736064629), the characteristic polynomial (2.6) reduces to

$$T(\lambda) = \lambda^2 - 1.9423140190465376\lambda + 1.$$

Then the roots of $T(\lambda) = 0$ are $\lambda_1 = 0.971157 - 0.238441i$ and $\lambda_2 = 0.971157 + 0.238441i$ with $|\lambda_1| = |\lambda_2| = 1$. Therefore, $(\alpha, \beta, h) = (0.01, 0.65, 0.6096575736064629) \in \Upsilon_{NS}$. The bifurcation diagrams and MLE are plotted in Fig. 7 for the system (1.3). Finally, for $\alpha = 0.01$, $\beta = 0.65$, $(x_0, y_0) = (0.65, 1.5)$ and various selected values of h in $[0.1, 1]$, the phase portraits of the system (1.3) are depicted in Fig. 8.

7 Concluding remarks

This work is related to qualitative analysis of discrete-time models describing the biochemical interaction in living cells in which these cells obtain energy by utilization of glucose. The discrete-time glycolysis models are obtained by Euler's method and nonstandard finite difference scheme. Implementation of linearized stability analysis for two-dimensional discrete-time systems gives explicit parametric conditions for dynamics of positive equilibria of these discrete-time glycolysis models. Moreover, it is proved that system (1.2) undergoes period-doubling bifurcation as well as Neimark–Sacker bifurcation over small range of bifurcation parameter h . Thus, the step size h for Euler's scheme has strong stability effects on positive steady-state or vice versa. It is investigated that there is no chance of emergence of period-doubling bifurcation for nonstandard discrete-time model (1.3). On the other hand, system (1.3) also endures Neimark–Sacker bifurcation as the bifurcation parameter h varies in small neighborhood of h_3 defined by

$$h_3 = \frac{\beta^2 - \alpha - \alpha^2 - 2\alpha\beta^2 - \beta^4}{\alpha^2 + 3\alpha\beta^2 + 2\beta^4}.$$

Furthermore, numerical simulation gives evidence of stable limit cycles for the system (1.3). Thus system (1.3) is more close to its continuous counterpart (1.1) with respect to emergence of limit cycles, whereas stronger evidence of emergence of bifurcation is found in model (1.2). Finally, two control strategies are implemented to control the chaos due to emergence of period-doubling and Neimark–Sacker bifurcations.

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